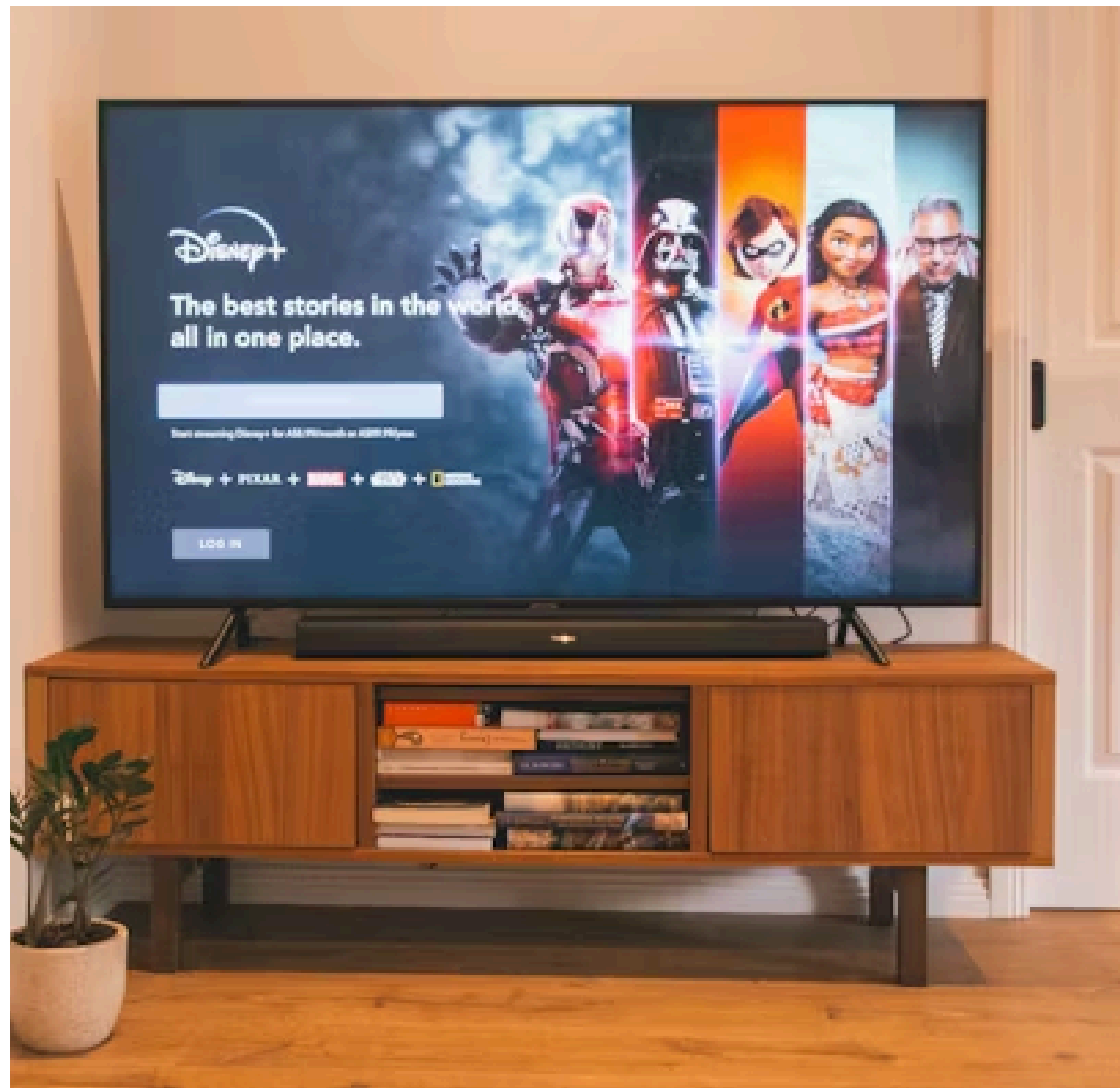




PROJECT 11

# SUBSCRIPTION SERVICE

Model Evaluation with Cross Validation

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PRESENTED BY MUNAWAR

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# BACKGROUND

In the era of subscription-based businesses, understanding and predicting customer spending is key to measuring Customer Lifetime Value (CLV). One of the most important financial metrics reflecting this accumulated value is **TotalCharges**—the total amount a customer has paid over their subscription period.

In business terms, the **TotalCharges** amount does not just happen, but is influenced by the dynamic interaction between several factors:

- Loyalty
- Regular Financial Commitments
- Contract & Service Conduct

Starting from this background, this notebook aims to build and evaluate two Ordinary Least Squares (OLS) regression models to predict customer **TotalCharges**. To ensure that the resulting model has good generalization capabilities and avoids overfitting, the evaluation process will use the 5-Fold Cross-Validation method with the R-squared metric as the main benchmark to see how much variance of TotalCharges can be explained by the features in the model.

# DATASET OVERVIEW

**973**

Total Customers

**17.5%**

Overall Churn Rate

**21**

Variables

**5-Fold**

Cross-Validation

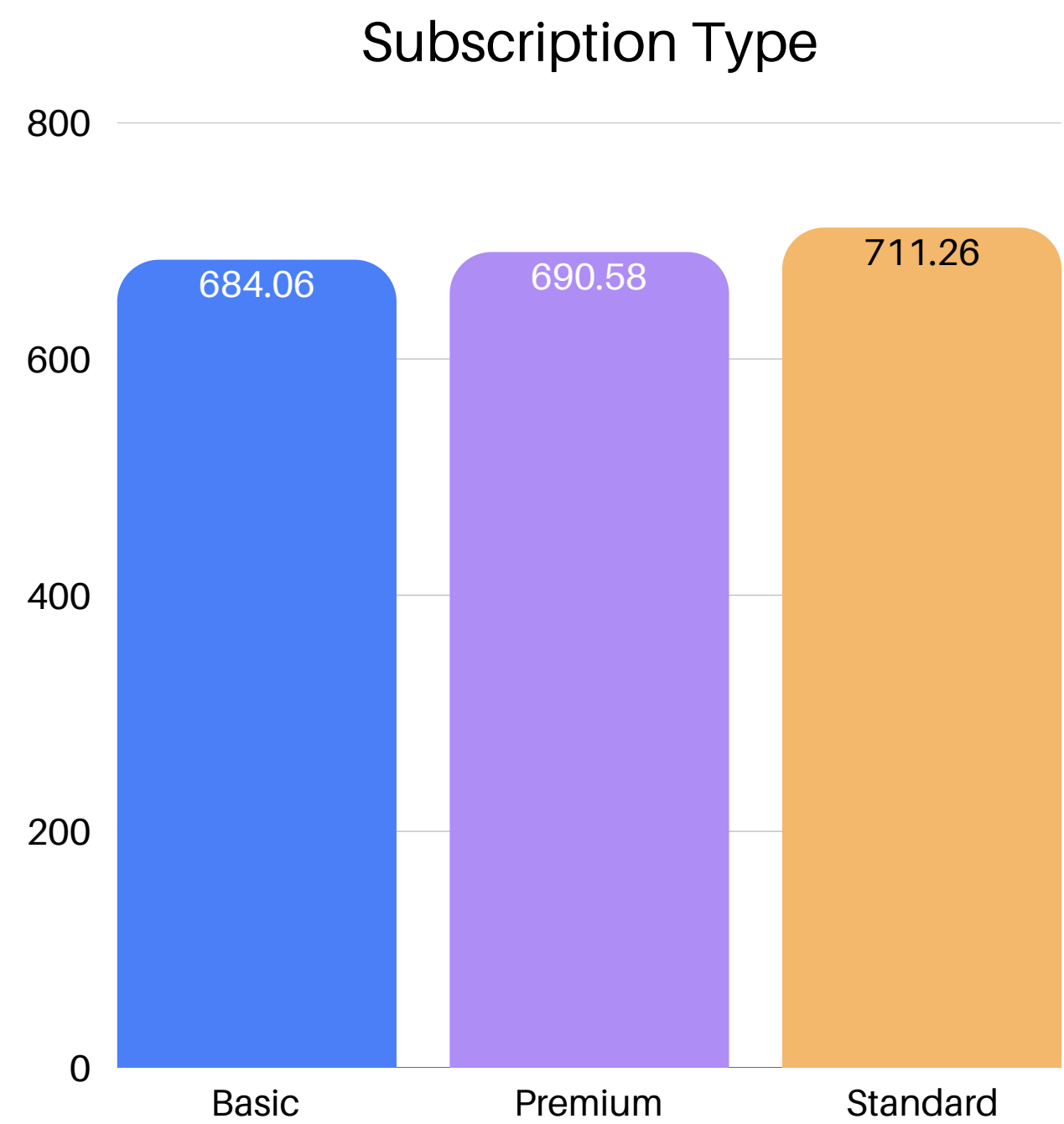
## About Dataset:

- Subscription types: Basic (30.4%), Premium (34.6%), Standard (29.8%)
- Payment methods: Bank transfer, Credit card, Electronic check, Mailed check
- Features include: AccountAge, MonthlyCharges, TotalCharges, ViewingHoursPerWeek, GenrePreference, and more
- Target variables: Churn (binary) and TotalCharges (regression)

# OBJECTIVE

- Predict TotalCharges using OLS regression with 5-Fold Cross-Validation ( $R^2$  metric). Two competing models were evaluated.
- Predict the expected total charges of a 25 month of account age and average viewing duration is 100/minutes

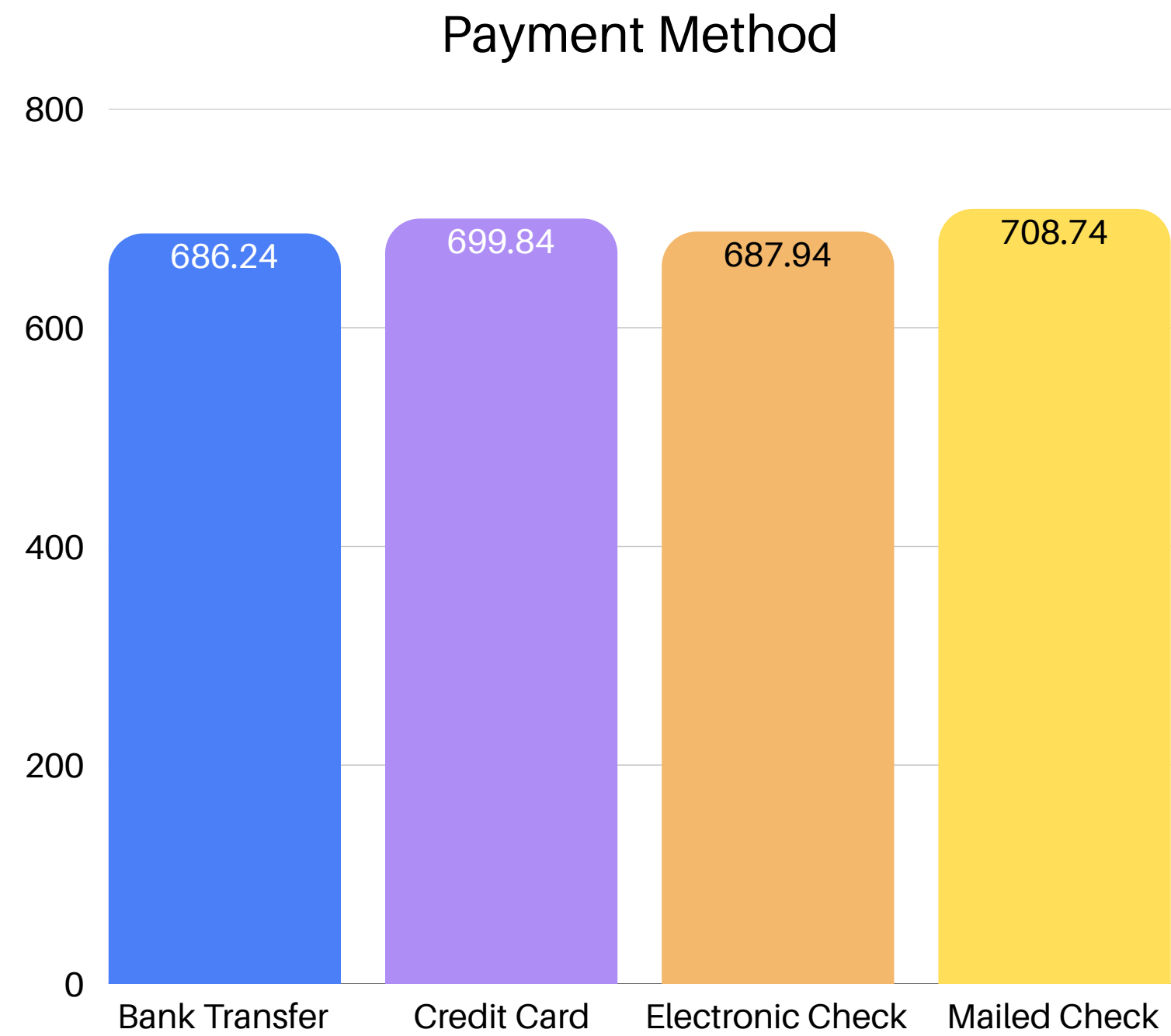
# EXPLORATORY DATA ANALYSIS (EDA)



**Key Insights:**

- The average total cost spent by users is in the range of 684 to 711.
- Users with the Standard subscription type turned out to have the highest average total expenditure (711.26), followed by Premium (690.58), and the lowest was Basic (684.06).

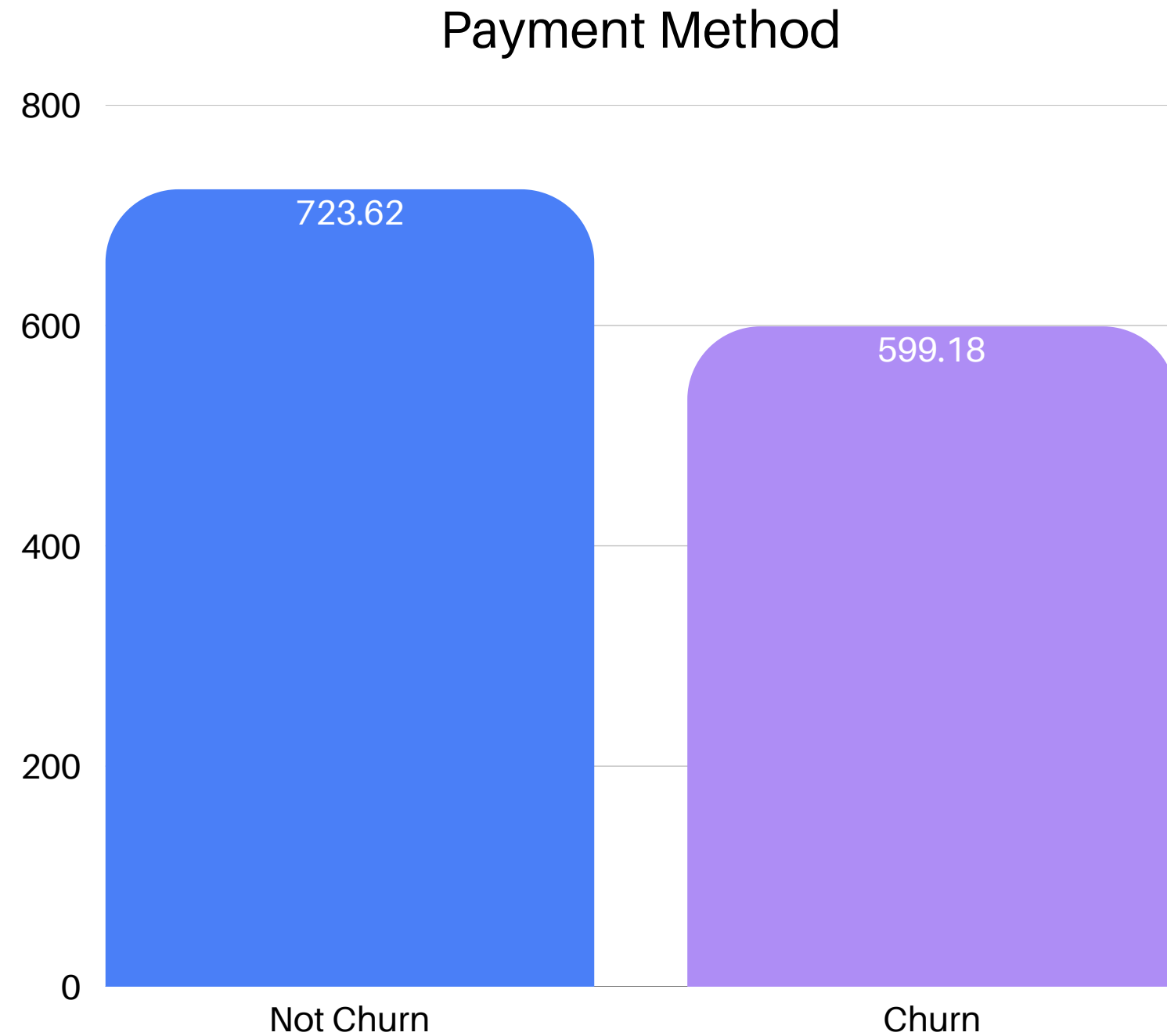
# EXPLORATORY DATA ANALYSIS (EDA)



## Key Insights:

- Based on payment method, the average expenditure ranges from 686 to 708.
- Customers using mailed checks had the highest average Total Charges (708.24), followed by credit cards (699.84). Bank transfers and electronic checks were close behind with similar figures (around 686–687).

# EXPLORATORY DATA ANALYSIS (EDA)



## Key Insights:

- There's a significant difference (around 124 points) between customers who stay and those who leave.
- Customers who churn have a significantly lower average total cost (TotalCharges) than those who remain loyal.

# OLS REGRESSION: MODEL DEFINITION

## Objective

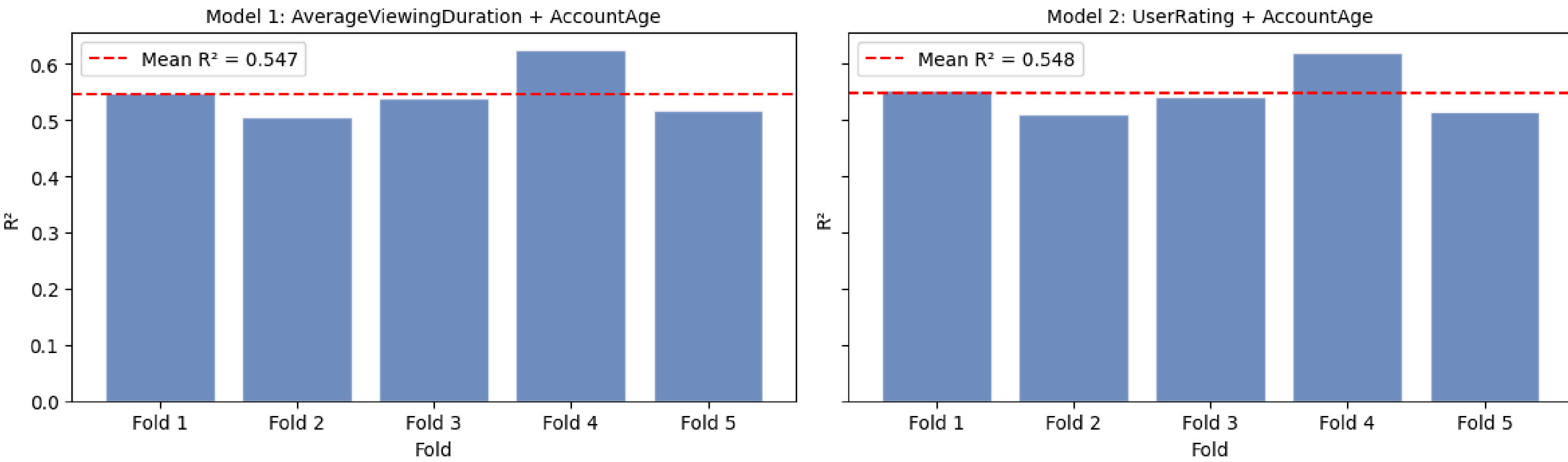
- Predict TotalCharges using OLS regression with 5-Fold Cross-Validation ( $R^2$  metric). Two competing models were evaluated.
- Predict the expected total charges of a 25 month of account age and average viewing duration is 100/minutes

| Model 1                                                                                                                                      |   |                        |   |
|----------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------|---|
| TotalCharges                                                                                                                                 | ~ | AverageViewingDuration | + |
| AccountAge                                                                                                                                   |   |                        |   |
| <ul style="list-style-type: none"><li>• Tests whether viewing behavior combined with account tenure explains subscription charges.</li></ul> |   |                        |   |

| Model 2                                                                                                                                              |  |
|------------------------------------------------------------------------------------------------------------------------------------------------------|--|
| TotalCharges ~ UserRating + AccountAge                                                                                                               |  |
| <ul style="list-style-type: none"><li>• Tests whether user satisfaction ratings combined with account tenure explain subscription charges.</li></ul> |  |

# 5-FOLD CROSS-VALIDATION RESULTS

5-Fold Cross-Validation R<sup>2</sup> Comparison



# 5-FOLD CROSS-VALIDATION RESULTS

## Model Comparison

| Model                               | Mean R <sup>2</sup> | Std R <sup>2</sup> |
|-------------------------------------|---------------------|--------------------|
| UserRating + AccountAge             | 0.547               | 0.044              |
| AverageViewingDuration + AccountAge | 0.546               | 0.046              |

- Both models achieve similar R<sup>2</sup> (~0.55), explaining ~55% of TotalCharges variance.
- Model 1 (AverageViewingDuration + AccountAge) edges out with Mean R<sup>2</sup> = 0.553.
- Low standard deviation in R<sup>2</sup> (~0.013) indicates consistent, stable generalization.
- AccountAge is the dominant predictor in both models (r = 0.746 with TotalCharges).

# FINAL MODEL: AVERAGEVIEWINGDURATION + ACCOUNTAGE

$$\text{TotalCharges} = \beta_0 + \beta_1 \times (\text{View Hours} - \overline{\text{View Hours}}) + \beta_2 \times \text{AccountAge}$$

| Intercept ( $\beta_0$ )                                                             | $\beta_1$ (c_view)                                                          | $\beta_2$ (AccountAge)                                                                  |
|-------------------------------------------------------------------------------------|-----------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>108.92</b>                                                                       | <b>- 0.32</b>                                                               | <b>10.18</b>                                                                            |
| Estimated TotalCharges for a customer at mean viewing duration and mean account age | Change in TotalCharges per 1 min deviation in viewing duration vs. the mean | Each additional month of account age adds ~\$10.18 to TotalCharges (dominant predictor) |

## Individual Prediction Example:

**Input:** AverageViewingDuration = 100 min, AccountAge = 25 months → **Predicted TotalCharges ≈ \$360.55** *(based on centered viewing + age-weighted regression)*

# CONCLUSION

1

Customers with longer account age tend to accumulate higher TotalCharges, indicating that retaining customers over time directly increases lifetime value. The subscription service should invest in long-term loyalty programs to extend account age.

2

The service can monitor AverageViewingDuration as a behavioral signal: customers with decreasing viewing duration may be at higher risk of disengagement and could be targeted with personalized content recommendations or retention offers.



## Limitation

Model explains ~55% of TotalCharges variance. Future models should incorporate SubscriptionType, PaymentMethod, and ContentPreference for improved explanatory power

**THANK YOU FOR  
YOUR TIME**



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**DATA ANALYST PRESENTATION**

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